nature portfolio

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Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our Editorial Policies and the Editorial Policy Checklist.

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed	
☐	☒	The exact sample size (*n*) for each experimental group/condition, given as a discrete number and unit of measurement
☐	☒	A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒	The statistical test(s) used AND whether they are one- or two-sided *Only common tests should be described solely by name; describe more complex techniques in the Methods section.*
☒	☐	A description of all covariates tested
☐	☒	A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒	A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☐	☒	For null hypothesis testing, the test statistic (e.g. *F*, *t*, *r*) with confidence intervals, effect sizes, degrees of freedom and *P* value noted *Give P values as exact values whenever suitable.*
☒	☐	For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐	For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☒	☐	Estimates of effect sizes (e.g. Cohen's *d*, Pearson's *r*), indicating how they were calculated

Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	qRT-PCR: StepOne Real-Time PCR software v2.3 Western Blot: Odyssey XF Imaging system (LICORbio) metabolomics: XCalibur (ThermoFisher) (immuno)histochemistry: Zeiss Zen 2.5 spectrophotometry: Synergy H1 (Biotek, BioSpx) flow cytometry: BD FACSDiva software version 6.1.3. sequencing data: Illumina Hiseq4000 platform and NovaSeq 6000 System Illumina statistics: GraphPad Prism 9
Data analysis	Western Blot: LICOR software metabolomics: XCalibur (ThermoFisher) image analysis: Zeiss ZEN 2.5; Zeiss AxioVision 4.9.1; ImageJ/FIJI 2.0.0; Qupath 0.5.0 spectrophotometry: SkanIt for Multiskan GO 2.0 flow cytometry: Kaluza software version 2.1 (Beckman Coulter) scRNA-seq analysis: CellRanger software (10x Genomics) and BIOMEX platform v1.0-5 statistics: GraphPad Prism 9 graphs: GraphPad Prism 9 figure preparation: Adobe Photoshop 2023

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio guidelines for submitting code & software for further information.

Data

Policy information about availability of data

All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our policy

Mouse scRNA-sequencing data have been deposited in the Array Express with accession code E-MTAB-14554. Source data for data represented in Figs. 1-7 and Extended Data Figs. 1-9 have been provided as Source Data files as wells as gel source images. All other data supporting the findings of this study and required to interpret, replicate or build upon the findings are available from the corresponding author upon reasonable request

Research involving human participants, their data, or biological material

Policy information about studies with human participants or human data. See also policy information about sex, gender (identity/presentation), and sexual orientation and race, ethnicity and racism.

Reporting on sex and gender	This study did not use human participants or human data
Reporting on race, ethnicity, or other socially relevant groupings	This study did not use human participants or human data
Population characteristics	This study did not use human participants or human data
Recruitment	This study did not use human participants or human data
Ethics oversight	This study did not use human participants or human data

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	In vitro sample sizes were based on previous similar studies that have given statistically significant results (doi: 10.1038/s41413-021-00185-7; doi: 10.1016/j.devcel.2020.05.001; doi: 10.1038/s41586-020-2050-1; doi: 10.1038/s41586-019-0874-3). For in vivo experiments, no statistical tests were used to pre-determine sample size, but sample size was chosen based on previous experiments and comparable studies in literature (doi: 10.1038/s41413-021-00185-7; doi: 10.1016/j.devcel.2020.05.001; doi: 10.1038/s41586-020-2050-1; doi: 10.1038/s41586-019-0874-3) and respects the limited use of animal models in line with the 3R recommendations: Replacement, Reduction, Refinement. Sample size for each experiment is indicated in the legend.
Data exclusions	No data were excluded
Replication	All experimental findings were reproduced in multiple independent experiments. For experiments using cultured cells, each datapoint represent chondrocytes isolated from a different mouse. For each figure panel, the number of independent experiments or biological replicates is indicated in the figure legends. Data shown in figure panels are the mean of all independent biological repeats. Western blot pictures are from representative experiments and the number of independent repeats is clearly indicated in the figure legends.
Randomization	We did not use statistical methods for randomization. For in vivo experiments, mice were randomly allocated into experimental groups. For in vitro experiments, cells were isolated from randomly chosen wild-type or conditional knockout mice.
Blinding	Blinding was used as much as possible in the study. Mice were given an unique number prior to data collection and analysis. Data collection and analysis, such as mouse phenotyping, immunostaining, qRT-PCR, Western blot and Flow analysis were frequently performed by participants other than the experiment designer. During these data collection and analysis steps, all participants were routinely blinded to group allocation. For in vitro experiments, investigators were blinded to group allocation during data collection and/or analysis by giving an unique identifier other that the sample name.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☐	☒ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☐	☒ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☐	☒ Flow cytometry
☒	☐ MRI-based neuroimaging

Antibodies

Antibodies used

Primary antibodies for Western blot analysis:
ATF4 (Cell Signaling Technology): D4B8 - #11815 - 1:1000 - Rabbit
ATF6 (Novus Biologicals): 70B1413.1 -#NBP1-40256SS - 1:1000 - Mouse
β-actin (Sigma-Aldrich): #A5441 - clone AC-15- lot 026M - 1:10000 - Mouse
eIF2a (Cell Signaling Technology): D7D3 - #5324 - 1:1000 - Rabbit
G6PDH (Cell Signaling Technology): polyclonal - #8866 - 1:1000 - Rabbit
p-eIF2a (Cell Signaling Technology): D9G8 - #3398 - 1:1000 - Rabbit
ubiquitin (Cell Signaling Technology): P4D1 - #3936 - 1:1000 - Mouse

Secondary antibodies for Western blot analysis:
anti-mouse HRP conjugated (Dako): polyclonal - P0161 - lot 00095192 - Rabbit
anti-rabbit HRP conjugated (Dako): polyclonal - P0448 - lot 00094764 - Goat

Primary antibodies for immunohistochemistry:
8-OHdG (Abcam): polyclonal - ab10802 - 1:500 - Goat
ATF4 (Cell Signaling Technology): D4B8 - #11815 - 1:20 - Rabbit
BrdU (Bio–Rad): BU1/75 (ICR1) - OBT0030 - 1:500 - Rat
COL2 (Chemicon, Sigma-Aldrich): CP18, II-4C11 - MAB8887 - 1:200 - Mouse
G6PDH (Abcam): EPR20668 - ab231828 - 1:500 - Rabbit

Secondary antibodies for immunohistochemistry:
anti-rabbit biotin conjugated (Dako): polyclonal - E0432 - 1:100 - Goat
anti-mouse FITC conjugated (Sigma-Aldrich): polyclonal - F0257 - 1:100 - Goat
anti-rat biotin conjugated (Dako): polyclonal - E0468 - 1:600 - Rabbit
anti-goat AF546 conjugated (Invitrogen): polyclonal - A21085 - 1:400- Rabbit

Amplification kit for biotin conjugated antibodies:
TSA Cyanine 3 System(Akoya Biosciences): NEL704A
TSA Fluorescein System (Akoya Biosciences): NEL701A

Primary antibodies for flow cytometry:
DNP (Sigma-Aldrich): 90451-S7150 - supplied in OxyBlot Protein Oxidation Detection kit

Secondary antibodies for flow cytometry:
anti-rabbit AF488 (ThermoFisher): polyclonal - A11008 - Goat

Validation

All antibodies were obtained from indicated commercial vendors with ensured quality and have been used in multiple experiments to detect intended proteins in control samples with expected molecular weight to validate their effectiveness in our study.
ATF4 (Cell Signaling Technology): validation by WB analysis as reported by provider
ATF6 (Novus Biologicals): validation by WB analysis as reported by provider
β-actin (Sigma-Aldrich): validation by WB analysis as reported by provider
eIF2a (Cell Signaling Technology): validation by WB analysis as reported by provider
G6PDH (Cell Signaling Technology): validation by WB analysis as reported by provider
p-eIF2a (Cell Signaling Technology): validation by WB analysis as reported by provider
ubiquitin (Cell Signaling Technology: validation by WB analysis as reported by provider
8-OHdG (Abcam): validation by immunohistochemical analysis as reported by provider
ATF4 (Cell Signaling Technology): validation by immunohistochemical analysis as reported by provider
BrdU (Bio–Rad): validation by immunocytochemical analysis as reported by provider
COL2 (Chemicon, Sigma-Aldrich): validated by immunocytochemical analysis on mouse growth plate and chondrocytes in culture
G6PDH (Abcam): validation by immunohistochemical analysis as reported by provider

Animals and other research organisms

Policy information about studies involving animals; ARRIVE guidelines recommended for reporting animal research, and Sex and Gender in Research

Laboratory animals	All mice that were used in this study were on a C57BL/6J genetic background and housed under conventional conditions at 22°C, 50-60% humidity, under a 14-hours light/10-hours dark cycle and fed a normal chow diet. Chondrocyte-specific G6PDH-deficient mice were generated by crossing G6PDH-floxed mice (harboring exon 10 flanked by LoxP sites; mice provided by P. Carmeliet, VIB-KU Leuven, Belgium) with transgenic mice expressing the Cre recombinase under the control of the collagen type II gene promotor (Ovchinnikov DA Genesis 2000). Wild-type littermates (G6PDH-floxed mice without expression of the Cre recombinase) were used as control. All mice were individually genotyped and correct excision of floxed alleles was determined. Mouse phenotyping was performed on 3-day-old mice. All colonies were housed and bred in individually ventilated cages in the animal facility of the KU Leuven.
Wild animals	This study did not use wild animals.
Reporting on sex	Phenotyping of chondrocyte-specific G6PDH KO mice showed that the phenotype was similar in male and female mice.
Field-collected samples	This study did not include field collected samples
Ethics oversight	All animal experiments were conducted according to the regulations and with approval of the Animal Ethics Committee of the KU Leuven, under the ECD number P254/2015 and P221/2017.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Plants

Seed stocks	*Report on the source of all seed stocks or other plant material used. If applicable, state the seed stock centre and catalogue number. If plant specimens were collected from the field, describe the collection location, date and sampling procedures.*
Novel plant genotypes	*Describe the methods by which all novel plant genotypes were produced. This includes those generated by transgenic approaches, gene editing, chemical/radiation-based mutagenesis and hybridization. For transgenic lines, describe the transformation method, the number of independent lines analyzed and the generation upon which experiments were performed. For gene-edited lines, describe the editor used, the endogenous sequence targeted for editing, the targeting guide RNA sequence (if applicable) and how the editor was applied.*
Authentication	*Describe any authentication procedures for each seed stock used or novel genotype generated. Describe any experiments used to assess the effect of a mutation and, where applicable, how potential secondary effects (e.g. second site T-DNA insertions, mosiacism, off-target gene editing) were examined.*

Flow Cytometry

Plots

Confirm that:

☒ The axis labels state the marker and fluorochrome used (e.g. CD4-FITC).

☒ The axis scales are clearly visible. Include numbers along axes only for bottom left plot of group (a 'group' is an analysis of identical markers).

☒ All plots are contour plots with outliers or pseudocolor plots.

☒ A numerical value for number of cells or percentage (with statistics) is provided.

Methodology

Sample preparation	Ex vivo flow cytometry (BrdU or BODIPY 581/591 C11): Chondrocytes were isolated from the growth plate using collagenase digest, 4 hours after IP injection of the fluorophore in 3-day-old mice Flow cytometry on in vitro cultured chondrocytes (BODIPY 591/591 C11, TPE-MI, CellMeter Pacific Blue, LysoTracker and SYTOX Green): Analysis was performed after incubation of cultured cells with the fluorophore, followed by detaching of the cells with collagenase. DNP flow cytometry on in vitro cultured chondrocytes : Chondrocytes were collected using collagenase detachment, fixed and permeabilized, followed by derivatization of carbonyl groups in DNP and antibody detection. For all flow cytometry experiments, unstained controls were included and positive controls were added where needed.

Instrument	BD FACSCantoII (BD Biosciensces)
Software	Data were collected using the BDFACSDiva software version 6.1.3 (BD Biosciences). Downstream data analysis was performed using Kaluza software version 2.1 (Beckman Coulter).
Cell population abundance	N/A as cells were not sorted, only processed for flow cytometry
Gating strategy	Cells were gated on FSC/SSC scatter plots. Subsequently, singlet cells were determined on a FSC-H/FSC-W scatter plot. Positivity was determined using unstained controls and mean fluorescent intensity was calculated where needed.

☒ Tick this box to confirm that a figure exemplifying the gating strategy is provided in the Supplementary Information.